

Chapter 5 Homework

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Exercise Group 5.2

- a) $\frac{1}{6} = 0.1\overline{6}$
- b) $\frac{5}{6} = 0.8\overline{3}$
- c) $\frac{122}{99} = 1.\overline{23}$
- d) $\frac{1}{13} = 0.\overline{076923}$
- e) $\frac{1}{7} = 0.\overline{142857}$
- f) $\frac{6}{5} = 1.2$
- g) $\frac{3}{2} = 1.5$

Exercise Group 5.3

- 2) Yes
- 3) Yes
- 4) No
- 5) A real number is any number that can be expressed by a finite number of digits preceding the decimal place followed by potentially infinite decimal digits (this is NOT the definition of the real numbers, but what I'm guessing the author is after). The rational numbers are all the real numbers that can be expressed in finitely many digits, or whose decimal part repeats. The irrational numbers are all the real numbers that are not rational.
- 6) Because the irrationals are defined as all the reals that aren't rational. So every real number is either rational or not rational i.e. irrational.
- 7) A fraction will terminate when written in decimal form iff the denominator's prime factorization only consists of 2s and 5s.

Exercise Group 5.4

- 1) $4.82 = 4 + \frac{8}{10} + \frac{2}{100}$; $1.29 = 1 + \frac{2}{10} + \frac{9}{100}$; $2.75 = 2 + \frac{7}{10} + \frac{5}{100}$; $7.122 = 7 + \frac{1}{10} + \frac{2}{100} + \frac{2}{1000}$;
 $0.8875 = \frac{8}{10} + \frac{8}{100} + \frac{7}{1000} + \frac{5}{10000}$; $2.40 = 2 + \frac{4}{10}$
- 2) 3.24, 2.356, 4.03, 1.1001
- 3) $0.\overline{6} = \frac{6}{10} + \frac{6}{100} + \frac{6}{1000} \dots$, $0.\overline{7} = \frac{7}{10} + \frac{7}{100} + \frac{7}{1000} \dots$, $0.\overline{12} = \frac{1}{10} + \frac{2}{100} + \frac{1}{1000} + \frac{2}{10000} \dots$
- 4) $1.\overline{2}$, $3.\overline{14}$, $0.\overline{8}$

Exercise Group 5.5

- 1)
 - a) If $3 > 2$ and $2 = 1$, then $3 > 1$.
 - b) If $3 > 2$ and $2 > 1$, then $3 > 1$.
 - e) $3 > 2$ iff $3 - 2 > 0$.
 - f) If $3 > 2$ and $1 \geq 0$ then $3 + 1 > 2 + 0$.
- 2) $a > b, a \geq b, u < w, a + c \leq r + s, (a + b)^2 > 0, z < 0$.
- 3) If x is a real number then $x^2 \geq 0$.

Proof. If x is a real number then $x^2 \geq 0$.

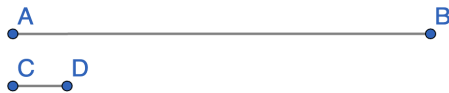
1. Let x be any real number, and observe that either $x > 0$, $x = 0$, or $x < 0$.
2. If $x > 0$, then $x^2 > 0$ since a positive times a positive is positive.
3. If $x < 0$, then $x^2 > 0$ since a negative times a negative is positive.
4. If $x = 0$, then $x^2 = 0$.
5. In either case, $x^2 \geq 0$.

□

- 4) Because a positive times a positive is positive.
- 5) Because a negative times a positive is negative.

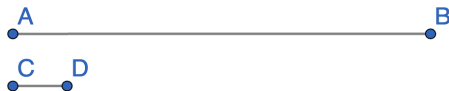
Exercise Group 5.6

- 1) 2
- 2) $n = 8$



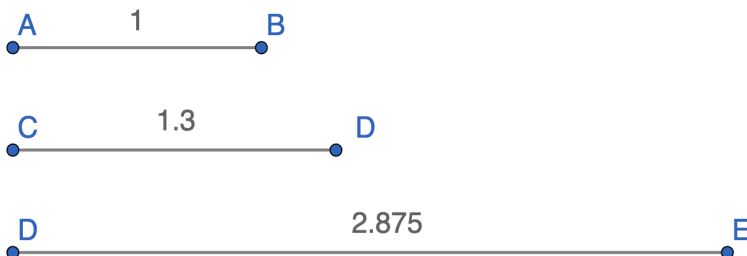
Exercise Group 5.7

- 1) The length is 7.5



Exercise Group 5.8

- 1) The following image is for both problems 1 and 2.



Exercise Group 5.9

- 1) $\overline{EF} = 7.26$ inches.
- 2) Yes.
- 3) Yes.
- 4) An isosceles triangle has two sides congruent, or an isosceles triangle has two sides of equal length.

Exercise Group 5.10

- 1) $\overline{AC} > 5$.
- 2) $\overline{AB} > 4$.
- 3) $\overline{AB} > \overline{EF}$.
- 4) $\overline{AD} > \overline{BC}$.
- 5) C is between A and B .

Exercise Group 5.11

- 1) $\overline{AB} = 5 \cdot 2.3 = 11.5$
- 2) The width of the room is $14 \cdot 12 = 168$ inches. This follows from Thm 5-3 because the unit of a foot has 12 units of an inch in it.
- 3) There are $1,000 \cdot 100 = 100,000$ centimeters in a kilometer.
- 4) $\overline{PQ} = 6.5 \cdot 0.5 = 3.25$
- 5) $\overline{AB}$ is a larger number when CD is the unit as opposed to when EF is the unit.

Algebra Review

- 1) $\overline{XT} = \frac{37}{2}$, and $\overline{TY} = \frac{19}{2}$.
- 2) $\overline{RS} = 33$.
- 3) If BAC , then $\overline{AB} = 6$ and $\overline{AC} = 18$. If ABC , then $\overline{AB} = 3$ and $\overline{AC} = 15$.
- 4) $\overline{AB} = 6$.
- 5)
 - a) The order is YXZ .
 - b) $\overline{XY} = \frac{2}{3}\overline{YZ}$.
 - c) If $\overline{XZ} - \overline{XY} = 6$, then $\overline{XY} = 12$.
- 6) $\overline{AB} = 32$.
- 7) *Proof.* Let $\overline{AC} = 2\overline{AB}$, $\overline{AB} + \overline{BC} = \overline{AC} + 2$. If $\overline{AB} > 1$, then A , B , and C are not colinear.
 1. Let $\overline{AB} > 1$, and we will show that it's impossible for ABC , BAC , and ACB , and so the points are not colinear.
 2. If ABC , then $\overline{AC} = \overline{AB} + \overline{BC}$ and $\overline{AB} + \overline{BC} = \overline{AC} + 2 \iff \overline{AC} = \overline{AB} + \overline{BC} - 2$.
 3. Then $\overline{AB} + \overline{BC} = \overline{AB} + \overline{BC} - 2 \iff 0 = -2 (\Rightarrow \Leftarrow)$
 4. Now if BAC , then $\overline{BC} = \overline{AB} + \overline{AC} \implies \overline{BC} = \overline{AB} + 2\overline{AB} \implies \overline{BC} = 3\overline{AB}$.
 5. Then $\overline{AB} + \overline{BC} = \overline{AC} + 2 \iff \overline{AB} + 3\overline{AB} = 2\overline{AB} + 2 \iff 4\overline{AB} = 2\overline{AB} + 2 \iff 2\overline{AB} = 2 \iff \overline{AB} = 1$, but it is already given that $\overline{AB} > 1 (\Rightarrow \Leftarrow)$.
 6. Now if ACB , then $\overline{AB} = \overline{AC} + \overline{BC} \implies \overline{AB} = 2\overline{AB} + \overline{BC} \implies 0 = \overline{AB} + \overline{BC}$.
 7. However, since $\overline{AB} > 1$ and all lengths are greater than or equal to zero, then $\overline{AB} + \overline{BC} > 0 (\Rightarrow \Leftarrow)$.
 8. Therefore A , B , and C are not colinear. □

8) $3(x + 6) = 4x \iff x = 18$. The perimeter of the triangle and square is 72 inches.

9) Let w be the width of the rectangle, l the length, and let s be the length of a side of the square. Then $2l + 2w + 4s = 36$, $l = 2w$, and $w = s + 1$. Then $2l + 2w + 4s = 36 \iff l + w + 2s = 18 \iff (2w) + w + 2s = 18 \iff 3w + 2s = 18 \iff 3(s + 1) + 2s = 18 \iff 3s + 3 + 2s = 18 \iff 5s + 3 = 18 \iff 5s = 15 \iff s = 3$. Then $s = 3$, $w = 4$, and $l = 8$. So the perimeter of the rectangle is $2l + 2w = 2(8) + 2(4) = 24$ ft. and the perimeter of the square is $4s = 4(3) = 12$ ft.

Answer: Rectangle is 24 feet; Square is 12 feet.

10) The point B is the midpoint of segment AC . Point X is between B and C , and $\overline{BX} = 3$ inches. Show that a square having a side congruent to AB has an area of 9 in more than a rectangle with sides congruent to AX and XC .

Proof. If B is the midpoint of AC , BXC , and $\overline{BX} = 3$ inches, then a square having side lengths $\overline{AB}$ has an area of 9 in² more than a rectangle with side lengths $\overline{AX}$ and $\overline{XC}$.

1. B is the midpoint of AC , BXC , and $\overline{BX} = 3$ inches.

2. Observe that $\overline{AB}^2$ is the area of a square having side lengths $\overline{AB}$, and $\overline{AXXC}$ is the area of a rectangle with side lengths $\overline{AX}$ and $\overline{XC}$.

3. Then $\overline{AB}^2 - \overline{AXXC} = \overline{AB}^2 - (\overline{AB} + 3)(\overline{AB} - 3) = \overline{AB}^2 - (\overline{AB}^2 + 3\overline{AB} - 3\overline{AB} - 9) = \overline{AB}^2 - (\overline{AB}^2 - 9) = \overline{AB}^2 - \overline{AB}^2 + 9 = 9$.

4. Therefore the difference in the areas is 9 in². $\square$

